

# Assignment 6

1. As quantum computing is known for exponential speed and for quantum parallelism world can store up to  $2^{10^21}$  quantum states due to superposition

2. Here  $|\Phi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$  is proved already.

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|1100\rangle + |0101\rangle + |0110\rangle - |1111\rangle)$$

We can assume  $|\Phi^+\rangle = |m\rangle \otimes |n\rangle$

$$|m\rangle = a_0|10\rangle + a_1|11\rangle \quad \& \quad |n\rangle = b_0|10\rangle + b_1|11\rangle$$

$$\text{Given } |\Phi^+\rangle = a_0b_0|1000\rangle + a_0b_1|1011\rangle + a_1b_0|1100\rangle + a_1b_1|1111\rangle$$

from  $a_0b_1 = 0$ , either  $a_0$  or  $b_1$  needs to be 0.

It means either  $a_0b_0 = 0$  or  $a_1b_1 = 0$  which contradicts

$a_0b_0 = 1$  or  $a_1b_1 = 1$ . So  $|\Phi^+\rangle$  is entangled.

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0100\rangle + |1101\rangle + |1110\rangle + |0111\rangle)$$

$$a_0b_0 = 0, \quad a_0b_1 = 1, \quad a_1b_0 = 1, \quad a_1b_1 = 0$$

We can assume  $a_0b_0 = 0$  either or them should be zero if denies our above calculation.

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} (|0100\rangle + |1101\rangle - |1110\rangle + |0111\rangle)$$

$$a_0b_0 = 0, \quad a_0b_1 = 1, \quad a_1b_0 = -1, \quad a_1b_1 = 0$$

We can assume  $a_0b_0 = 0$  and one of them should be zero if denies calculation.

## Assignment 6 (10)

3. Yes I believe entanglement is the concept irrespective of distance we can expect the result. It will be profitable for him.

4. Here as mentioned basis states are orthonormal and if we have the same state in both two spaces then the inner product will be 1.

5. Given  $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$

$$P = \frac{1}{2} |0\rangle\langle 0| + \frac{1}{2} |0\rangle\langle 1| + \frac{1}{2} |1\rangle\langle 0| + \frac{1}{2} |1\rangle\langle 1|$$

$$P = \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix}$$

$$P = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

As we know entanglement entropy is zero because our given state is not product of two subspaces.

6. Given  $P = \frac{1}{2} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

We know  $P_B = \text{Tr}_A(P)$

For tracing second qubit (trace the B space)

$$= \begin{pmatrix} \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ \boxed{0} & \boxed{\frac{1}{2}} & \boxed{-\frac{1}{2}} & \boxed{0} \\ \boxed{0} & \boxed{-\frac{1}{2}} & \boxed{\frac{1}{2}} & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{pmatrix}$$

Above same shape elements can be added from above tracing B space

$$= \begin{bmatrix} 0 & x & 0 & x \\ x & \frac{1}{2} & x & 0 \\ 0 & x & \frac{1}{2} & x \\ x & 0 & x & 0 \end{bmatrix}$$

$$\rho_A = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$= -\text{Tr}_A (\rho_A \log_2 (\rho_A))$$

$$= -\text{Tr}_A (\rho_A \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix})$$

$$= -\text{Tr}_A \left( \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \right) = 1$$

7.  $U_{\text{NOT}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$$U^\dagger U = I$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

8. CNOT is  $U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

$$U^\dagger U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \times 1 + 0 \times 0 + 0 \times 0 + 0 \times 0 & 0 & 0 & 0 \\ 0 & 0 \times 0 + 1 \times 1 + 0 \times 0 + 0 \times 0 & 0 & 0 \\ 0 & 0 & 0 \times 0 + 1 \times 1 + 0 \times 0 + 0 \times 0 & 0 \\ 0 & 0 & 0 & 0 \times 0 + 1 \times 1 + 0 \times 0 + 0 \times 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = I$$